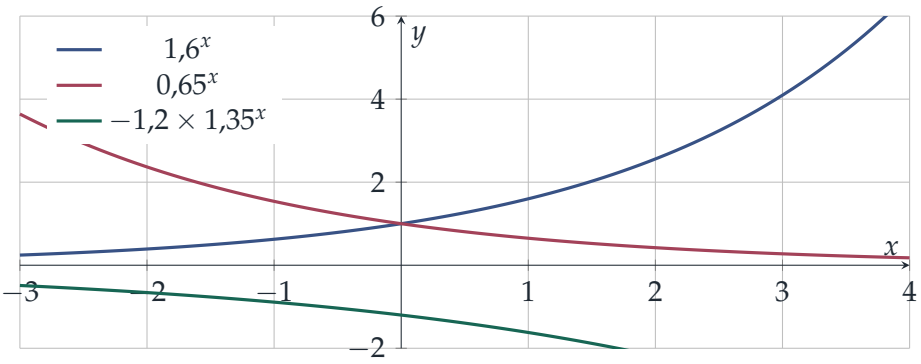
Chapter 2

Exponential functions

Competences: model a continuous evolution, change register, calculate, interpret

2.1 Reference course

For $a > 0$, the function $x \mapsto a^x$ extends to all real the entire powers of a. It checks $a^0 = 1$ and $a^{-x} = \dfrac{1}{a^x}$.



Key ideas

If $a > 1$, $x \mapsto a^x$ is strictly increasing. If $0 < a < 1$, it is strictly decreasing. $f(x) = ka^x$, multiply by $k < 0$ reverses the direction of variation. In any case, $a^x > 0$.

For $a > 0$ and all real x, y :

$$a^{x+y} = a^x a^y, \qquad a^{x-y} = \frac{a^x}{a^y}, \qquad (a^x)^y = a^{xy}.$$

Frequent error

In general, $a^x + a^y \neq a^{x+y}$. The properties relate to products and quotients, not to sums.

2.1.1 Developments and average rate

An initial value quantity V_0 subject to a constant relative change in coefficient q modeled by $V(t) = V_0 q^t$. If an overall coefficient Q the result of n the same trends, the average coefficient is $q = Q^{\frac{1}{n}}$ and average rate is $t = q - 1$.

Worked example

A turnover increased from 80,000 to 95,000 euros in four years.
$Q = \dfrac{95\,000}{80\,000} = 1{,}1875$. The average annual coefficient is $q = 1{,}1875^{\frac{1}{4}} \approx 1{,}0440$: the average annual rate

So it's about 4.40%.

2.1.2 Solve and interpret

When both members can be written on the same basis, the subjectivity is used: $a > 0$ and $a \neq 1$, $a^u = a^v$ is equivalent to $u = v$. Otherwise, a graphical or numerical resolution gives a value approached; the decimal logarithm, then studied, will give a formula.

Session 7 — Definition and direction of variation a^x

Exponential functions

06/10/2026 • B : 2 h – C : 1 h

Objectives of the session

· Give meaning to a^x.

· Base linkage and variation.

· Read curves.

Fluency · 6 to 8 minutes, without calculator

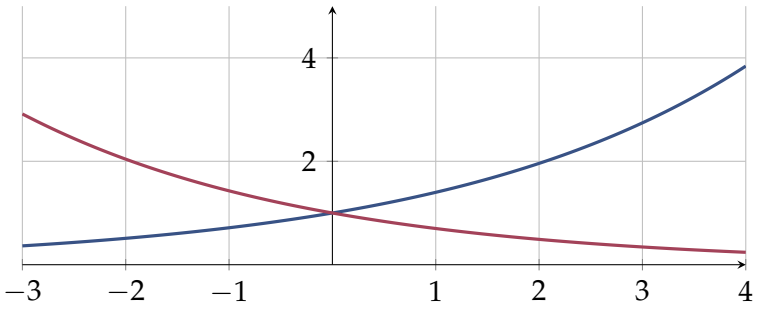
1. Write the multiplier coefficient of a 10 per cent increase . . . . . .
2. Compare 1.04^3 and 1.04^7. . . . . .
3. Calculate $2^4 \times 2^3$. . . . . .
4. Write 5^{-2} in fraction form . . . . . .
5. The function $x \mapsto 0{,}7^x$ Is it increasing? . . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, have the keywords reformulated: give meaning to a^x and link base and variation.

Research activity

Using a calculator, complete a table of values of 1 $,5^x$ and $0{,}7^x$ for $x = -2, -1, 0, 1, 2, 3$.
Trace points, conjecture variations and identify the common point of both curves.



Trace written

For $a > 0$, a^x is defined for all real x. The function is increasing if $a > 1$ and decreasing if $0 < a < 1$.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Training · essential

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation in $f(x) = 4 \times 1{,}04^x$ and $g(x) = -4 \times 0{,}8^x$.
3. An initial value of 1350 increases by 4% per year. Give the model and calculate the value After six years.

Reference in the manual

Supplements: Calao, courses p. 42·47; exercises p. 48, n° 1 to 8.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 48, exercise 2

On 1 ^{1st} January 2019, France counts 66 ,It is estimated that 9 million people live in the region. Annual at 0,4 per cent for all natural n, we note u_n the number of millions of inhabitants in France on 1 ^{1st} January of year 2019 $+ n$ and we admit that the rest (u_n) is a geometric sequence.

1. Give common difference / ratio for follow-up (u_n). Express u_n according to the n.
2. Propose expression of a function f to estimate the number of inhabitants in France x years after 1 ^{1st} January 2019, x is a real number. Determine, according to this model, the Number of inhabitants per 1^{1st} January 2022, on 1^{1st} July 2023 and 1 ^{1st} October 2025.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 8 — Algebraic power and properties

Exponential functions

08/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Transform powers.

· Use negative exhibitors.

· Identify an inapplicable rule.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 4 - x + 2$. .....
2. Calculate $2^1 \times 2^3$. .....
3. Write 5 $^{-2}$ in fraction form .....
4. The function $x \mapsto 0{,}7^x$ Is it increasing? .....
5. Compare 1.04 ^{3} and 1.04^7. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before activity, reformulate keywords: transform powers and use exhibitors negative.

Research activity

Simplify 2 $^3 \times 2^{-5}$, $5^{x+2} \div 5^x$ and $(3^x)^2$. Then invent a numerical counter example to $a^x + a^y = a^{x+y}$.

Trace written

$a^{x+y} = a^x a^y$, $a^{x-y} = \dfrac{a^x}{a^y}$ and $(a^x)^y = a^{xy}$. These rules require the same positive basis.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Simplify 3 $^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation in $f(x) = 5 \times 1{,}05^x$ and $g(x) = -5 \times 0{,}8^x$.
3. An initial value of 1400 increases by 5% per year. Give the model and calculate the value After six years.

Reference in the manual

Supplements: Calao, courses p. 43·45; exercises p. 49, n° 9-13.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 49, exercise 10

Write in the form of a single power of 2:

$$a = 2^5 \times 2^3; \qquad b = \left(2^{4,5}\right)^3; \qquad c = 2^3 \times 4^2; \qquad d = \frac{2^5}{4}.$$

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 9 — Functions $x \mapsto ka^x$: variations and graphs

Exponential functions

13/10/2026 • B : 2 h – C : 1 h

Objectives of the session

· Study ka^x.

· Combine sign and variation.

· Formula and graphic link.

Fluency · 6 to 8 minutes, without calculator

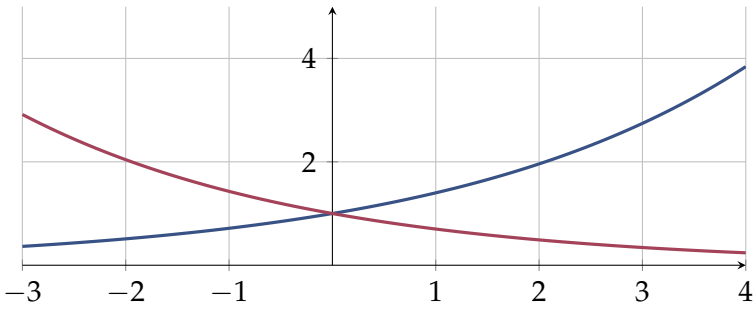
1. Calculate 20 per cent of 100 . . . . . .
2. Write 5^{-2} in fraction form . . . . . .
3. The function $x \mapsto 0{,}7^x$ Is it increasing? . . . . . .
4. Compare 1.04^3 and 1.04^7. . . . . .
5. Calculate $2^2 \times 2^3$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: study ka^x and combine sign and variation.

Research activity

Associate the four functions 2^x, $0{,}6^x$, -2^x and $-0{,}6^x$ with four curves. Justify each choice by the sign, the point of order originally and the direction of variation.



Trace written

For $f(x) = ka^x$, the sign is that of k. If $k < 0$, multiplication by k reverses the direction of variation of a^x.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Simplify $3^{x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation in $f(x) = 6 \times 1{,}06^x$ and $g(x) = -6 \times 0{,}8^x$.
3. An initial value of 1450 increases by 6% per year. Give the model and calculate the value After six years.

Reference in the manual

Supplements: Calao, courses p. 44·47; exercises p. 48, n° 5 to 8, and p. 52, n° 31 to 36. reproduced in this session is identified below by its page and number; the reference allows also find the figures and digital resources of the manual.

Manuel Calao · p. 48, exercise 6

Determine the direction of variation of the functions defined on $\mathbb{R}$ by:

$$a : t \mapsto -2 \times 1{,}4^t ; \qquad b : t \mapsto 9{,}85 \times 0{,}95^t ; \qquad c : t \mapsto 0{,}8 \times 2{,}25^t .$$

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 10 — Average rate of change

Exponential functions

15/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Calculate an average rate.

· Distinguish global and average rate.

· Interpret an exponent $\frac{1}{n}$.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 25 per cent increase . . . . . .
2. The function $x \mapsto 0{,}7^x$ Is it increasing? . . . . . .
3. Compare 1.04^3 and 1.04^7. . . . . .
4. Calculate $2^3 \times 2^3$. . . . . .
5. Write 5^{-2} in fraction form . . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, have the keywords reworded: calculate an average rate and distinguish overall rate and Medium.

Research activity

The price of a service increases from 18 to 23,50 euros in five years. Calculate the overall coefficient, then the Average annual coefficient and average annual rate. Check by recomposing five times the evolution.

Trace written

If Q is the overall coefficient of n the same trends, the average coefficient is $q = Q^{\frac{1}{n}}$ and rate average is $q - 1$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Simplify $3^{x+2} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation in $f(x) = 2 \times 1{,}02^x$ and $g(x) = -2 \times 0{,}8^x$.
3. An initial value of 1500 increases by 2 % per year. Give the model and calculate the value After six years.

Reference in the manual

Supplements: Calao, courses p. 47·49; exercises p. 49, n. 14-19, and p. 54, n. 47-50.
reproduced in this session is identified below by its page and number; the reference allows
also find the figures and digital resources of the manual.

Manuel Calao · p. 49, exercise 18

The turnover of a company has increased by 38% over three years.
annual average to within 0.1%.

Personal work

Resuming Exercise 3 with the following data:
170 and the rate or common difference / ratio by 6. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and
indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 11 — Reading graphical representations

Exponential functions

03/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Read images and preimages.

· Compare two models.

· To estimate graphically.

Fluency · 6 to 8 minutes, without calculator

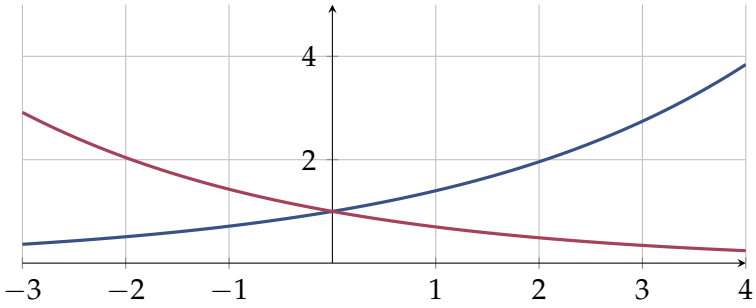
1. Reduce 3 $x + 7 - x + 2$. .....
2. Compare 1.04 ^{3} and 1.04^{7}. .....
3. Calculate $2^{4} \times 2^{3}$. .....
4. Write 5 $^{-2}$ in fraction form .....
5. The function $x \mapsto 0{,}7^{x}$ Is it increasing? .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before activity, reformulate keywords: read images and preimages and compare two models.

Research activity

Two commercial offers are modelled by $f(x) = 120 \times 1{,}08^{x}$ and $g(x) = 180 \times 0{,}96^{x}$. Trace curves on $[0, 12]$, read their intersection point and interpret the areas where each value is the Larger.



Trace written

Read $f(x)$ consists of from x to the curve and then to the axis of the y-orders.
of y consists of from y towards the curve and then towards the axis of the apses.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Simplify $3^{x+3} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation in $f(x) = 3 \times 1{,}03^x$ and $g(x) = -3 \times 0{,}8^x$.
3. An initial value of 1550 increases by 3% per year. Give the model and calculate the value After six years.

Reference in the manual

Supplements: Calao, pp. 52, Nos. 31 to 36. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 52, exercise 38

The following functions were represented: f and g defined by: $f(x) = 2{,}9^x$ and $g(x) = 0{,}35^x$.

1. What is the representative curve of f? To justify this choice.
2. Graphically solve the equation $f(x) = 4$. Using the graphic calculator allows to obtain a solution close to within 0.01.
3. To determine graphically all the reals x such as: $f(x) \leqslant 4$ and $g(x) \leqslant 4$.

Personal work

Resuming Exercise 2 with the following data:
175 and the rate or common difference / ratio by 7. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 12 — Simple exponential equations

Exponential functions

05/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Solve with the same base.

· Approach a solution.

· Check the likelihood.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 160 .....
2. Calculate $2^1 \times 2^3$. .....
3. Write 5^{-2} in fraction form .....
4. The function $x \mapsto 0{,}7^x$ Is it increasing? .....
5. Compare 1.04^3 and 1.04^7. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: solve with the same basis and approach a solution.

Research activity

Solve $4^x = 64$, $9^x = \frac{1}{3}$ and $1{,}12^x = 2$: first on the same basis where possible, then numerically for the last. Frame the last solution between two integers.

Trace written

To solve $a^x = b$: use the same base when possible; otherwise frame or use a digital tool. Check that $b > 0$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation in $f(x) = 4 \times 1{,}04^x$ and $g(x) = -4 \times 0{,}8^x$.
3. An initial value of 1600 increases by 4% per year. Give the model and calculate the value After six years.

Reference in the manual

Supplements: Calao, p. 52, No. 37·38, then p. 54, No. 45·46. The exercise reproduced in this sitting is identified below by its page and number; the reference also allows to find the figures and numerical resources of the manual.

Manuel Calao · p. 52, exercise 33

Solve on $\mathbb{R}$ Iinquation: $0.8^{x} \geqslant 0{,}8^{3}$.

Personal work

Resuming Exercise 3 with the following data:
180 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 13 — Modeling by an exponential function

Exponential functions

10/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Build an exponential model.

· Identify parameters.

· Provide and interpret.

Fluency · 6 to 8 minutes, without calculator

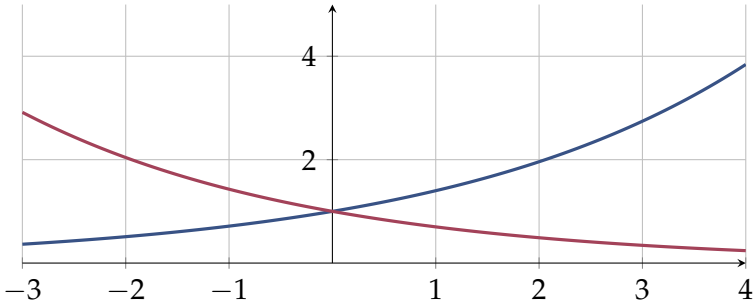
1. Write the multiplier coefficient of a 10 per cent increase . . . . . .
2. Write 5^{-2} in fraction form . . . . . .
3. The function $x \mapsto 0{,}7^x$ Is it increasing? . . . . . .
4. Compare 1.04^3 and 1.04^7. . . . . .
5. Calculate $2^2 \times 2^3$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: build an exponential model and identify the parameters.

Research activity

An application has 12,000 users at launch and its attendance increases by 9% by months. Build $f(t)$, calculate $f(6)$, then determine numerically when it will exceed 25,000 users.



Trace written

A constant relative change in coefficient q modeled by $f(t) = V_0 q^t$. V_0 is the value initial and q the coefficient per unit of time.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Simplify $3^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation in $f(x) = 5 \times 1{,}05^x$ and $g(x) = -5 \times 0{,}8^x$.
3. An initial value of 1650 increases by 5% per year. Give the model and calculate the value After six years.

Reference in the manual

Supplements: Calao, p. 50, No. 20-25, and p. 53, No. 41.
identified below by its page and number; the reference also allows you to find the figures and the manual's digital resources.

Manuel Calao · p. 50, exercise 25

The atmospheric pressure is 1 013 hPa (hectoPascal) at sea level, and decreases by 12 % each time we climb by 1 000 m. This is an exponential decline.
It can be modeled by a function P of altitude h in thousands of meters checking: $P(h) = ka^h$.

1. Determine constants k and a.
2. Calculate the pressure at an altitude of 5,500 m at 1 hPa.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 14 — Assessment and evaluation of the chapter

Exponential functions

12/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Mobilize the achievements of the chapter.

· Write a model.

· Correct an error.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 3 - x + 2$. .....
2. The function $x \mapsto 0{,}7^x$ Is it increasing? .....
3. Compare 1.04^3 and 1.04^7. .....
4. Calculate $2^4 \times 2^3$. .....
5. Write 5^{-2} in fraction form .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: mobilising the achievements of the chapter and drafting a modeling.

Research activity

From four student productions, identify and correct: confusion rate/coefficient, property of false powers, error of sense of variation and conclusion without unity.

Trace written

To validate an exponential model: identify V_0 and q, calculate with the units, check the order of and interpret the result.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Simplify $3^{x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation in $f(x) = 6 \times 1{,}06^x$ and $g(x) = -6 \times 0{,}8^x$.
3. An initial value of 1700 increases by 6% per year. Give the model and calculate the value After six years.

Reference in the manual

Supplements: Calao, pp. 54, Nos. 47 to 58. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 55, exercise 59

1. A reduction of 20% a year, then 80% a year. What is the percentage average annual decline?

2. What is the average rate of change corresponding to an increase of 60%, followed by a decrease 60%?

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.